Total Marks: 30

Section A: Multiple-Choice Questions (15 Marks)

Q1. Choose the correct answer.

1. What is the primary purpose of a computer network?
   a) To increase computer storage
   b) To share resources and exchange data
   c) To replace operating systems
   d) To convert data into images

2. Which Cisco Packet Tracer mode allows packets to be observed step by step?
   a) Realtime Mode
   b) Simulation Mode
   c) Physical Mode
   d) Configuration Mode

3. How many bits are in an IPv4 address?
   a) 16
   b) 32
   c) 64
   d) 128

Section B: Scenario-Based Questions (15 Marks)

Q2. Subnet Design and Binary Conversion

A university has been assigned the network 192.168.20.0/24 for two departments:

- **Data Science:** Requires 50 host addresses.
- **Administration:** Requires 25 host addresses.

Answer the following questions using VLSM.

a) Convert the IP address 192.168.20.70 into binary, showing all four octets.

b) Design suitable subnets for both departments, allocating the larger subnet first.

c) For each department, calculate the following:

- Network ID
- Subnet mask
- First and last usable IP addresses
- Broadcast address

Q3. Cisco Packet Tracer Implementation and Viva

Part A: Practical Implementation

A university wants to connect its Data Science and Administration LANs through a router so that computers in both departments can communicate.

Part B: Viva (5 marks)
